

Human-AI Collaboration for Co-Managing Epilepsy in Real Time

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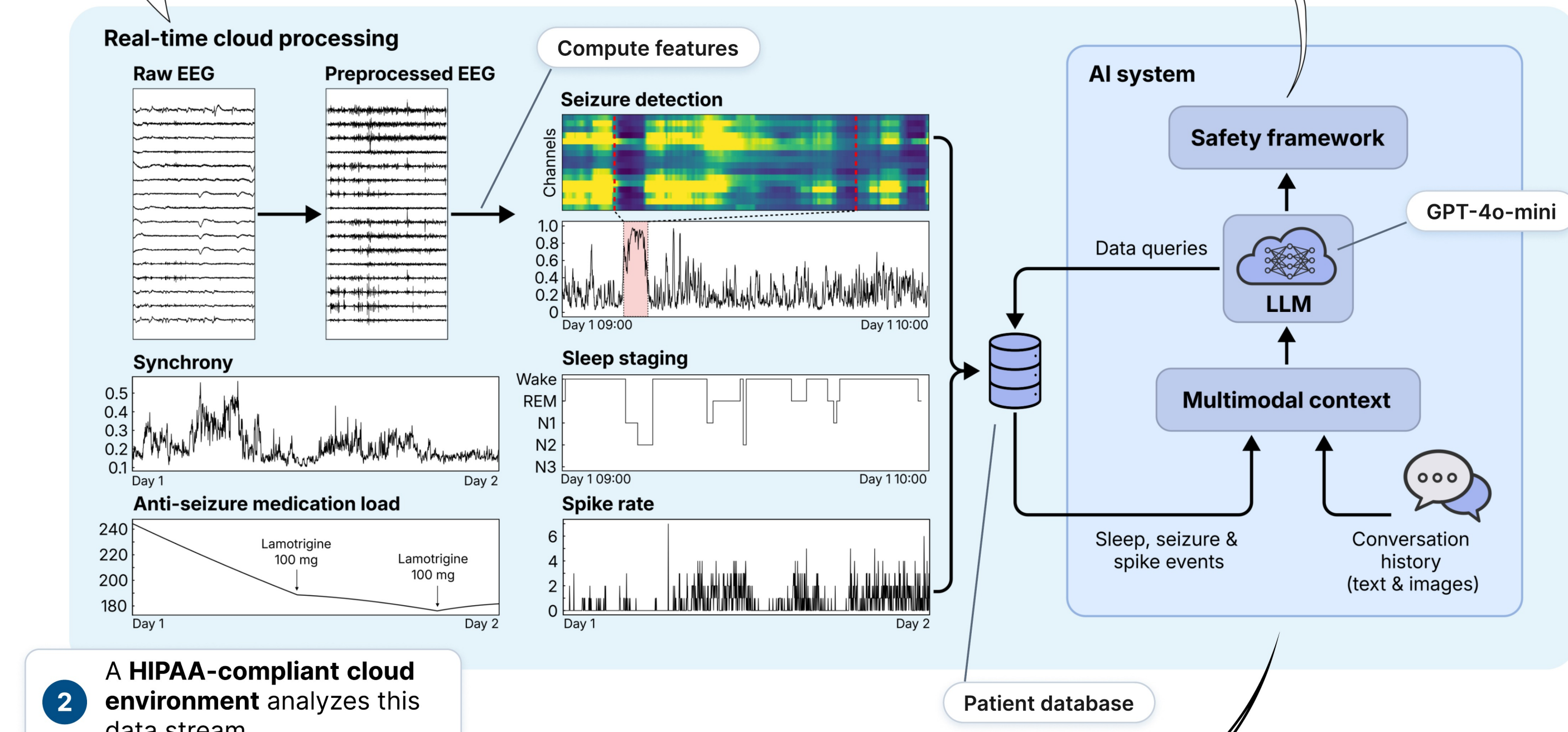
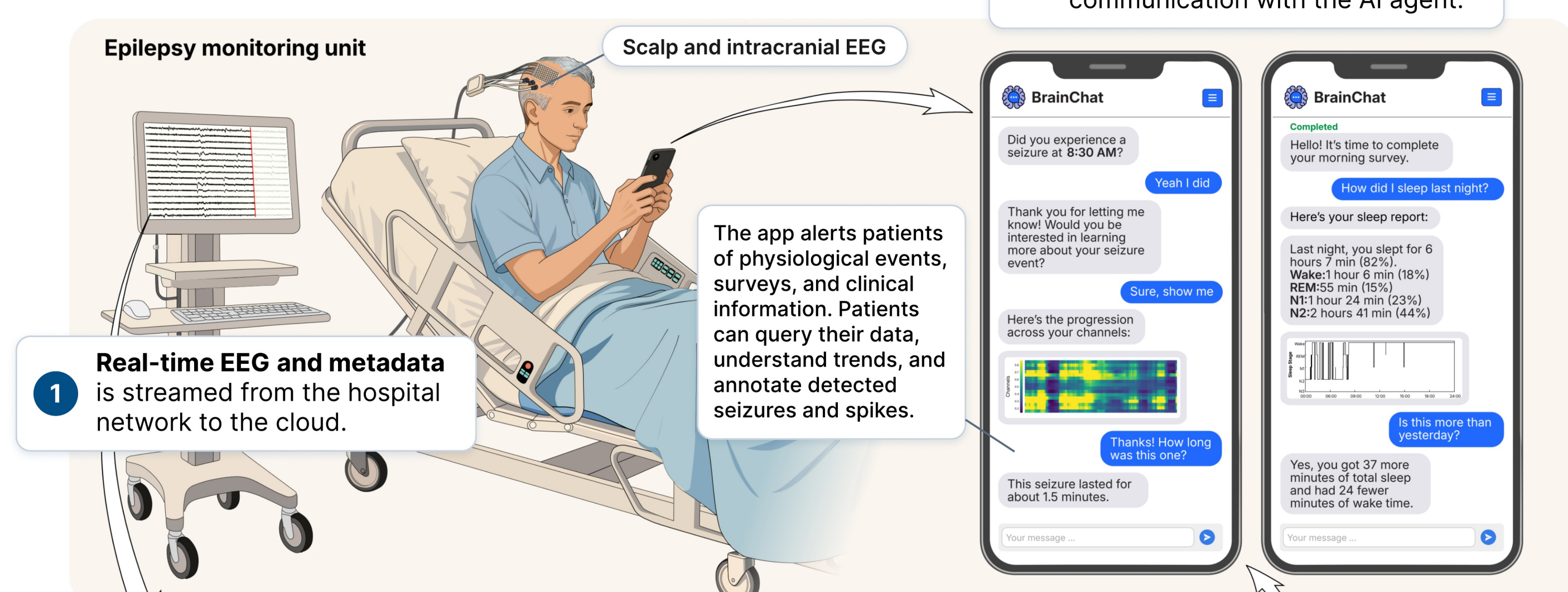
Background

- Over one-third of the world's **70 million people with epilepsy** have seizures not controlled by medication¹.
- Implantable devices can reduce seizures and detect biomarkers of seizure risk, yet none share this information with patients in real-time or learn from their feedback².
- Here, we present a novel platform that interprets physiological activity and behavior in real-time and allows **two-way, spontaneous communication between patients and their devices** to co-manage disease.
- We report results from a prospective pilot study of **13 patients** in the epilepsy monitoring unit (EMU) at the Hospital of the University of Pennsylvania.

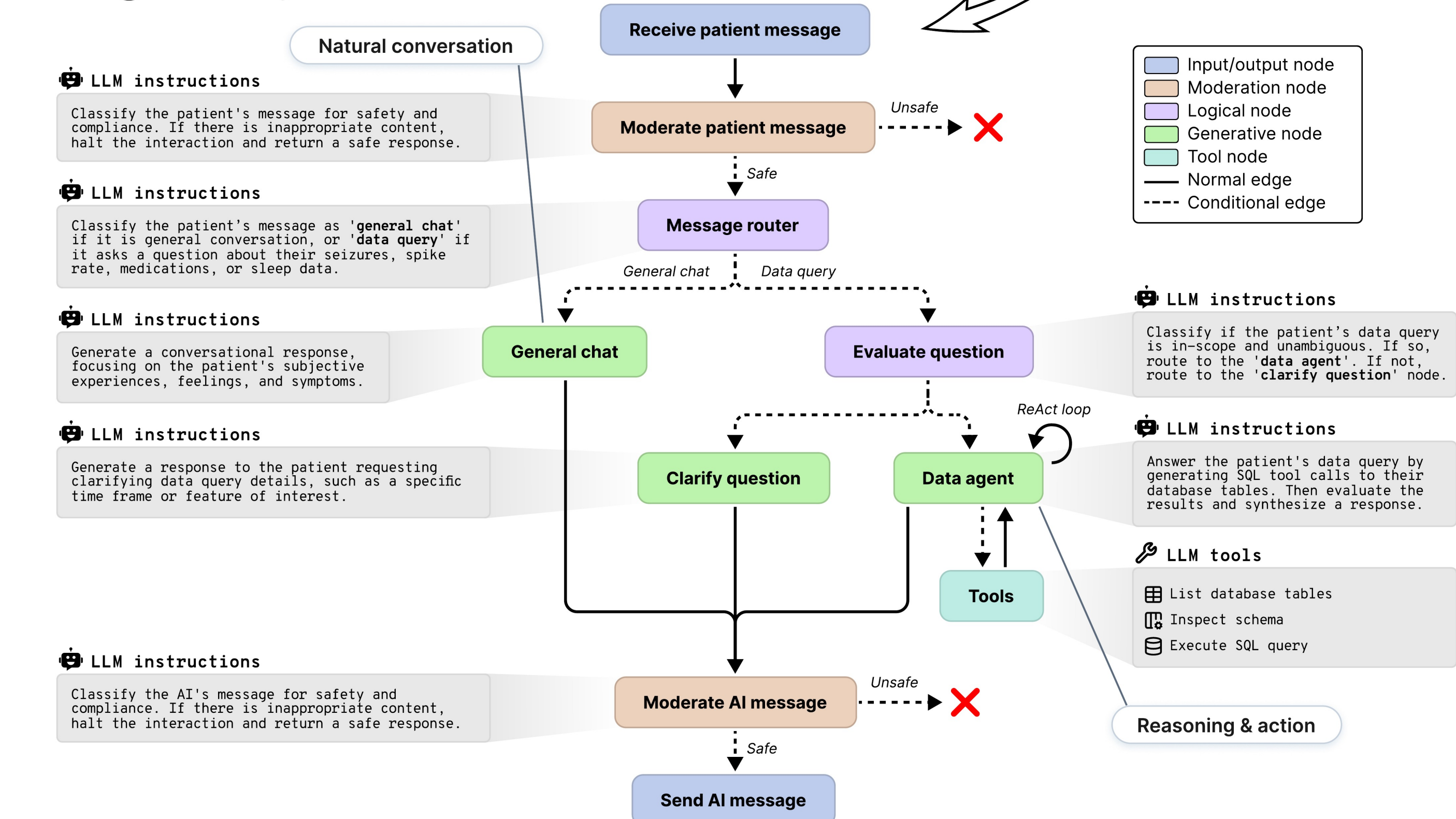
Methods

Patient-AI interface in the EMU

The platform comprises **three** systems:

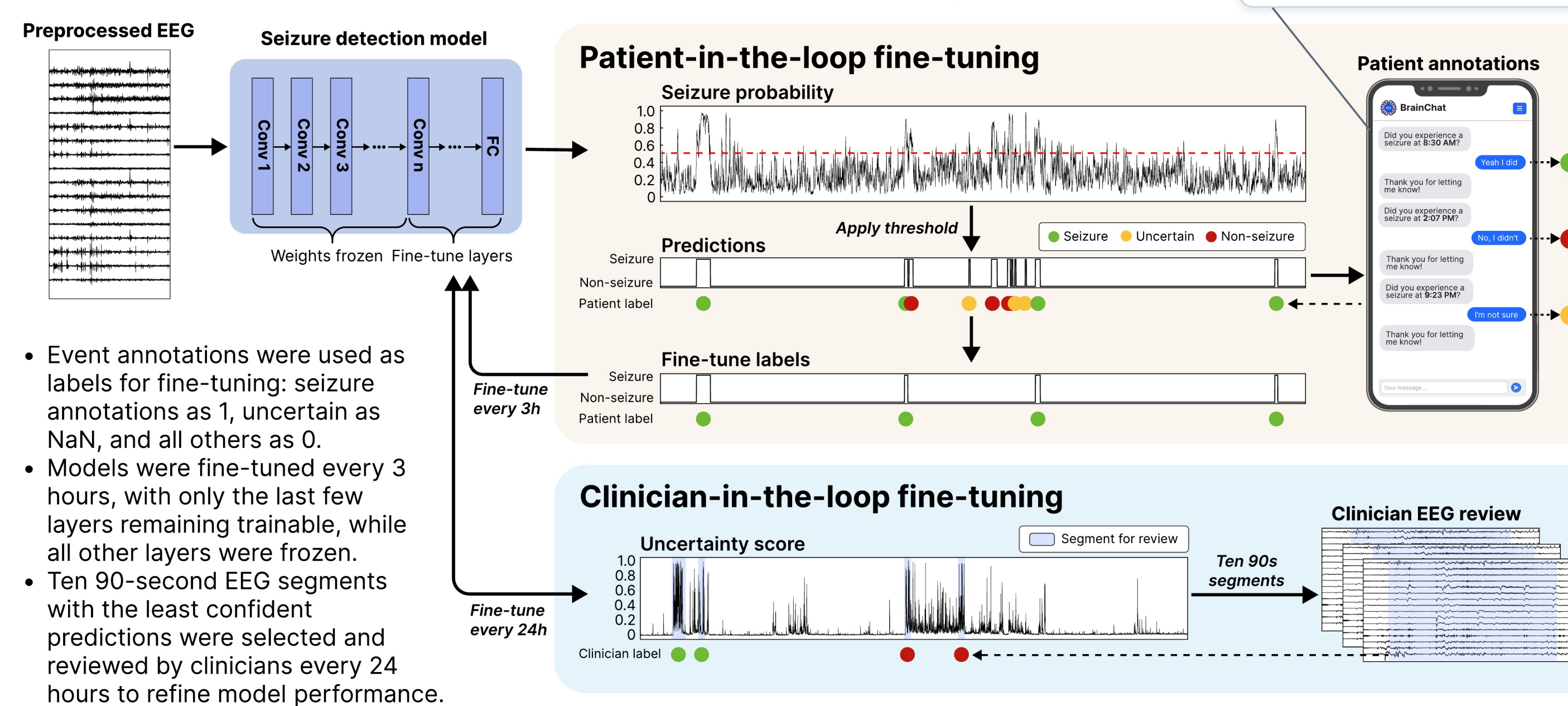


AI agent system



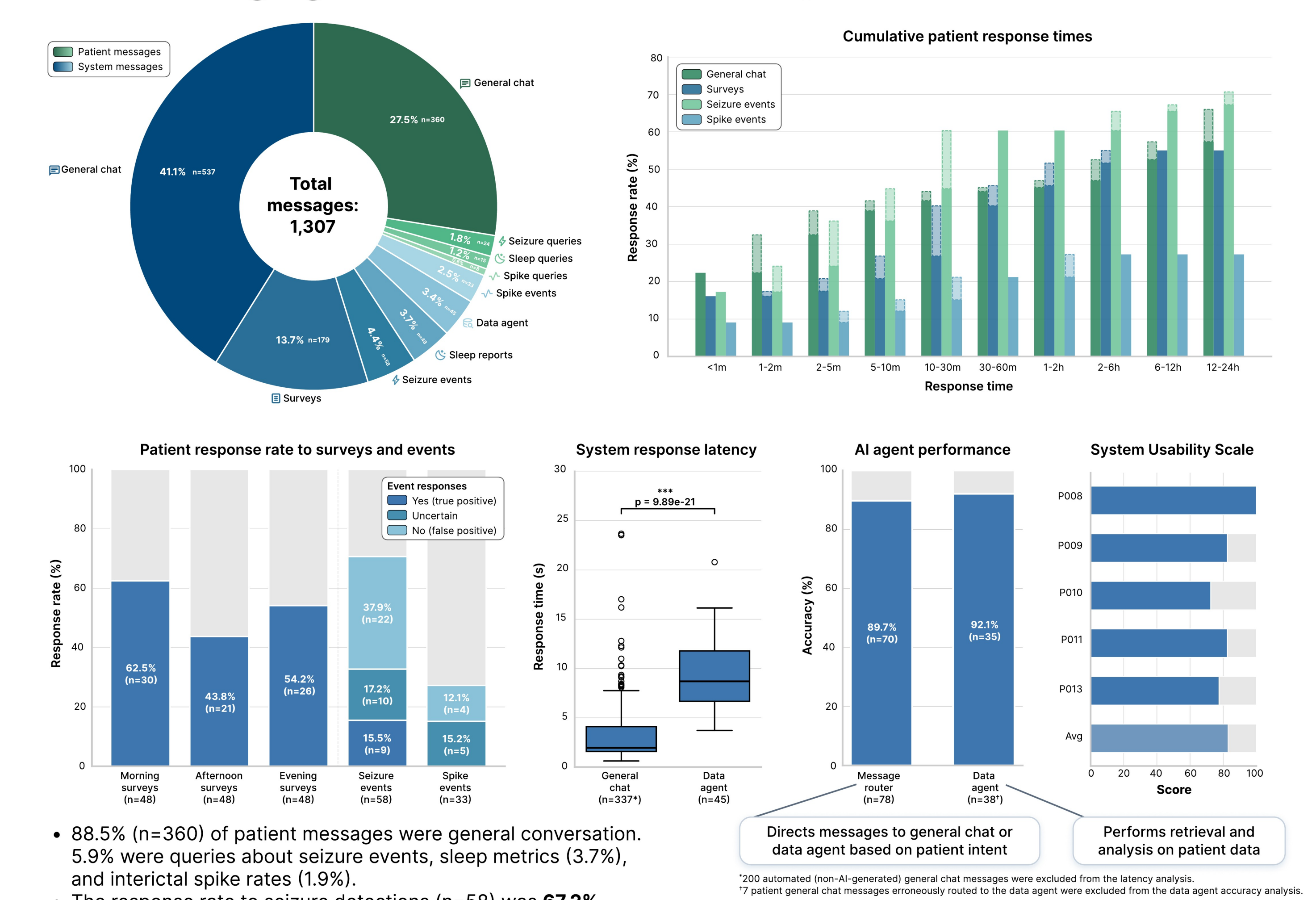
Seizure detection fine-tuning

We propose that fine-tuning using real-time patient responses to seizure detections can improve model performance without requiring clinician input.



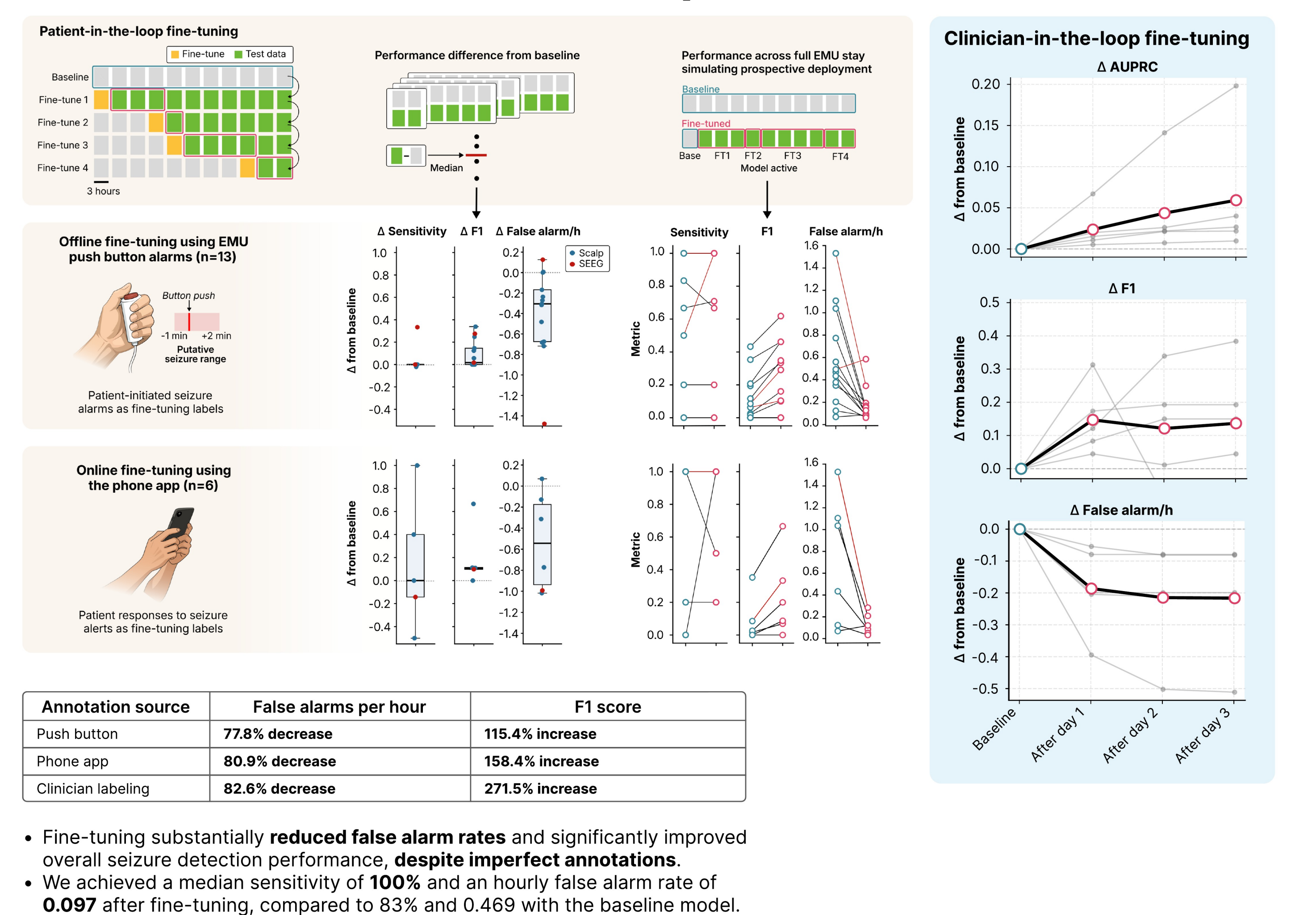
Results

Patient engagement & AI technical performance



- 88.5% (n=360) of patient messages were general conversation. 5.9% were queries about seizure events, sleep metrics (3.7%), and interictal spike rates (1.9%).
- The response rate to seizure detections (n=58) was **67.2%**.
- The system flagged **1.47% (6/407)** of patient messages as inappropriate or unsafe. **0.0% (0/900)** of the LLM-generated messages were flagged. This was corroborated by human review.

Patient and clinician labels improve seizure detection



Conclusion

- We enabled patients to communicate with their monitoring devices, utilizing inpatient EEG monitoring equipment with cloud-based compute, storage, and LLMs in a HIPAA-secure environment.
- Patients could initiate conversations, receive alerts and surveys, and ask questions to query their data through a smartphone application.
- The system learned and re-trained itself to markedly improve seizure detection performance using imperfect patient annotations alone, without the need for clinician intervention.
- This approach is broadly adaptable beyond epilepsy: from Parkinson's DBS programming to cardiac devices, insulin pumps, and neuropsych disorders.
- Future work will extend this platform to outpatient settings, incorporate additional biosensors, and optimize therapeutic stimulation.

References and acknowledgments

- Löschner, W. et al. Drug resistance in epilepsy: clinical impact, potential mechanisms, and new innovative treatment options. *Pharmacol Rev.* Jul;72(3):606-638 (2020).
- Nair, D. R. et al. Nine-year prospective efficacy and safety of brain-responsive neurostimulation for focal epilepsy. *Neurology* 95 (2020).

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Read our pre-print!



Soon to be published in
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